

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester MCA (CBCS) Degree Examinations
 December 2024- January 2025

22MCA301 – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from each **Unit**.

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) What is Artificial Intelligence? Discuss the various branches of Artificial Intelligence.	10	L*2	1	1
	b) Identify and elaborate the challenges associated with achieving the goals of AI.	10	L2	1	1
2.	a) List types of artificial intelligence. Describe the characteristics of each type.	10	L2	1	1
	b) Discuss the advantages and disadvantages of using AI.	10	L1	1	1
Unit – II					
3.	a) What is reinforcement learning? Discuss the concepts of reinforcement learning problem and its characteristics.	10	L2	2	1
	b) Explain the concept of supervised learning with an example.	10	L2	2	1
4.	a) How does reinforcement learning differ from supervised and unsupervised learning? Illustrate with an example.	10	L2	2	1
	b) Discuss briefly the applications of Machine Learning.	10	L2	2	1
Unit – III					
5.	a) Write and explain ID3 algorithm to construct a decision tree.	10	L2	3	1
	b) Discuss the significance of prior, likelihood, and posterior in the context of Bayes' Theorem.	10	L2	3	1
6.	a) Briefly discuss the difference between linear regression and logistic regression with examples.	10	L2	3	1
	b) Define polynomial regression. Explain how does it differ from linear regression.	10	L2	3	1
Unit – IV					
7.	a) Explain the K-Nearest Neighbors (KNN) algorithm. What are its strengths and weaknesses?	10	L2	4	1
	b) Write a note on Principal Component Analysis.	10	L2	4	1
8.	a) Define K-means clustering. Describe its purpose and general application in data analysis.	10	L2	4	1
	b) Explain the difference between agglomerative and divisive hierarchical clustering with illustrations.	10	L2	4	1
Unit – V					
9.	a) Define Neural Network. Discuss the applications of Neural Network.	10	L2	5	1
	b) Describe Gradient Descent algorithm with an example.	10	L2	5	1
10.	a) Describe the backpropagation process in deep learning, and explain how it helps in training a neural network.	10	L2	5	1
	b) Briefly discuss the layers in Artificial Neural Network. Explain how the neural network works.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
